

Exercises Hydrological Modelling

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1 Introduction

This book contains exercises for the course [Hydrological Modelling](#) at Ghent University. The focus of this set of exercises will be on rainfall-runoff modelling with a conceptual hydrological model.

The objective of the practical exercises of this course is to illustrate some of the most important steps necessary to perform a proper modelling of a medium-scale catchment in Flanders. Only some specific steps of the process will be illustrated here across 5 exercises:

1. Delineation of the catchment (Chapter [4](#))
2. Implementation of a hydrological model (Chapter [5](#))
3. Calibration of the hydrological model (Chapter [6](#))
4. The analysis of uncertainty in the model output of the model (Chapter [7](#))
5. Assimilation of observations into the model (Chapter [8](#)).

Needless to say that for each step, there are many options, and that the choice for e.g. a specific model, calibration algorithm, or evaluation criterium depends on the specific situation, the aim of the modelling exercise, and the personal taste of the modeller. For an extensive – yet non-exhaustive – overview of different techniques in hydrological modelling, we refer to the theoretical course notes and specialized literature.

The aim of the practicals is to:

1. Make you acquainted with the advantages and disadvantages of the selected algorithms,
2. Make you aware of the potential problems related to modelling rainfall-runoff relationships, and
3. Point you to and make you reflect on existing alternatives for the algorithms selected here.

2 Practical information

2.1 Working with Quarto

2.2 Evaluation

3 Study area and data

For these exercises, a hydrological dataset collected in the Zwalm catchment will be used. The Zwalm catchment is located in the center of Belgium and can be considered as a medium-scale catchment.

4 Exercise 1: Delineation of the Zwalm catchment

5 Exercise 2: Implementation of the model

6 Exercise 3: Calibration of the model

7 Exercise 4: Uncertainty analysis of the model output

8 Exercise 5: Data assimilation

References